1	**(a)**	(gravitational) potential at infinity defined as/is zero	B1	
		(gravitational) force <u>attractive</u> so work got out/done as object moves from infinity (so potential is negative)	B1	[2]
	(b) (i)	$\Delta E = m\Delta\phi$ $= 180 \times (14 - 10) \times 10^{8}$	C1	
		$= 7.2 \times 10^{10}$ J	A1	
		increase	B1	[3]
	(ii)	energy required = $180 \times (10 - 4.4) \times 10^{8}$ *or* energy per unit mass = $(10 - 4.4) \times 10^{8}$	C1	
		$\frac{1}{2} \times 180 \times v^{2} = 180 \times (10 - 4.4) \times 10^{8}$ *or* $\frac{1}{2} \times v^{2} = (10 - 4.4) \times 10^{8}$	C1	
		$v = 3.3 \times 10^{4}$ m s^{-1}	A1	[3]